

Ahmed Khaled

Software Engineer

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ABOUT

Cairo-based senior computer science student and software engineer with hands-on development experience across corporate environments, teams, internships and competitive hackathons. Focused on writing clean, maintainable code to build functional software systems and reliable multi-tenant applications.

EDUCATION

- **Ain Shams University** Cairo, Egypt
Bachelor of Science in Computer Science; GPA: 3.3 2022 – 2026

EXPERIENCE

- **Fawry** Smart Village, Egypt
Full-Stack Software Engineer Intern 2025
 - Worked with Java and Spring Boot in an enterprise backend environment
 - Practised software design patterns, RESTful API design, and database management
 - Collaborated with cross-functional teams following Agile/Scrum methodologies
- **Banque Misr** Cairo, Egypt
Software QA/QC Engineer Intern Summer 2024
 - Automated regression test suites using Cypress, reducing manual testing effort across key banking workflows
 - Collaborated within an Agile/Scrum team to triage bugs and validate fixes against functional and non-functional requirements
 - Managed test cases, execution cycles, and defect tracking via Jira and Zephyr Scale
- **Commercial International Bank (CIB)** Cairo, Egypt
Internship Trainee Summer 2024
 - Explored digital transformation initiatives, data analytics, data literacy, and cybersecurity practices
 - Gained practical knowledge of core banking operations and modern financial services technology

PROJECTS

- **Sawtak — Graduation Project / Senior Thesis** Ain Shams University
Anonymous civic whistleblowing platform with cryptographic privacy guarantees 2026
 - Architected a four-layer trust-zone system: public internet → privacy proxy (DMZ) → internal backend → Custom POA blockchain
 - Privacy proxy layer strips all identifying headers and PII, manages anonymous sessions via Redis, and enforces sliding-window rate limits
 - Anonymous evidence uploaded to IPFS/Pinata with CIDs committed on-chain; identified complaints encrypted with AES-256-GCM in PostgreSQL with private Cloudflare R2 storage
 - AI content filtering before any database or chain write
 - 184 unit/route-layer tests and 24 integration tests; full Prometheus + Grafana observability stack with alerting
- **AgencyKit — B2B SaaS Startup Venture** Proprietary / In Development
All-in-One Content Operations Platform for Digital Agencies 2026
 - Architected a multi-tenant content operation engine featuring logically isolated client portals, dedicated team dashboards, and secure asset workspaces.
 - Engineered a structured state-machine flow governing the full asset lifecycle: Brief → Design Iteration → Multi-Stage Internal Review → Client Approval Workflow.

- Integrated AI driven campaign planning modules alongside version-controlled iteration trackers for graphic and text assets.
- Designed decoupled publishing queues interfacing directly with major social network graph webhooks (Meta, X, LinkedIn, TikTok, YouTube).

- **Rabet — InnovEgypt Startup (ITIDA / TIEC)** Cairo, Egypt
OBD-II hardware + software module with B2B insurance integration 2025
 - Designed system architecture integrating an OBD-II hardware module with a mobile app for real-time vehicle health checks, fault diagnostics, and emergency alerts
 - Engineered a "black box" logging mechanism to securely store impact telemetry, critical diagnostic trouble codes (DTCs), and vehicle velocity vectors upon collision detection
 - Integrated continuous dashcam video feed capturing with automated event-triggered snapshot isolation during erratic driving behaviors or telemetry anomalies
 - Structured a B2B commercial model bundling telemetry cloud bridging services with insurance policy products — awarded Certificate of Acknowledgement C0505S25T0677
- **repAI — Google GDG Cairo AI Finance Hackathon** Google Developer Groups Cairo
AI-powered recycling platform with cashback rewards — Ranked 4th place 2025
 - Real-time object detection using Google Gemini to instantly identify, categorize, and evaluate recyclable materials
 - Built an automated valuation engine assigning real-time, market-based value based on scanned items' material and size
 - Designed the cashback flow enabling seamless wallet accumulation and reward redemption via eco-partner physical branches
- **TL;DR — NASA Space Apps Challenge 2025** NASA Space Apps
RAG-powered research summarization and knowledge engine for space biology 2025
 - Engineered a context-aware RAG system digesting 2,000+ technical documents to deliver tailored research summaries
 - Implemented an interactive conversational AI assistant generating multilingual voice summaries in English, Arabic, and French
 - Optimized backend retrieval latency to seamlessly query over 608 summarized peer-reviewed space biology papers

EXTRACURRICULAR ACTIVITIES

- **Open Source Community FCIS** Cairo, Egypt
Senior Web Committee Staff Member 2023 – 2025
 - Delivered technical sessions and lectures on modern web development topics
 - Contributed to the community website built on the MERN stack, collaborating across multiple sub-teams

SKILLS

- **Programming Languages:** JavaScript, TypeScript, Go, Python, Java, SQL, C, C++, PHP, Dart
- **Frameworks & Runtimes:** React.js, Next.js, Bun, ElysiaJS, Node.js, Angular, Tailwind CSS, Bootstrap, Flutter
- **Databases & Cloud:** PostgreSQL, Redis, Convex, Cloudflare R2, Google Cloud Run, Docker/Podman
- **Testing & QA:** Cypress, Zephyr Scale, Selenium, Manual Testing (Test Cases, HLS)
- **Tools:** Git, GitHub/GitLab, Jira, Figma, WordPress
- **Languages (Spoken):** English (Native / Bilingual), Arabic (Native)